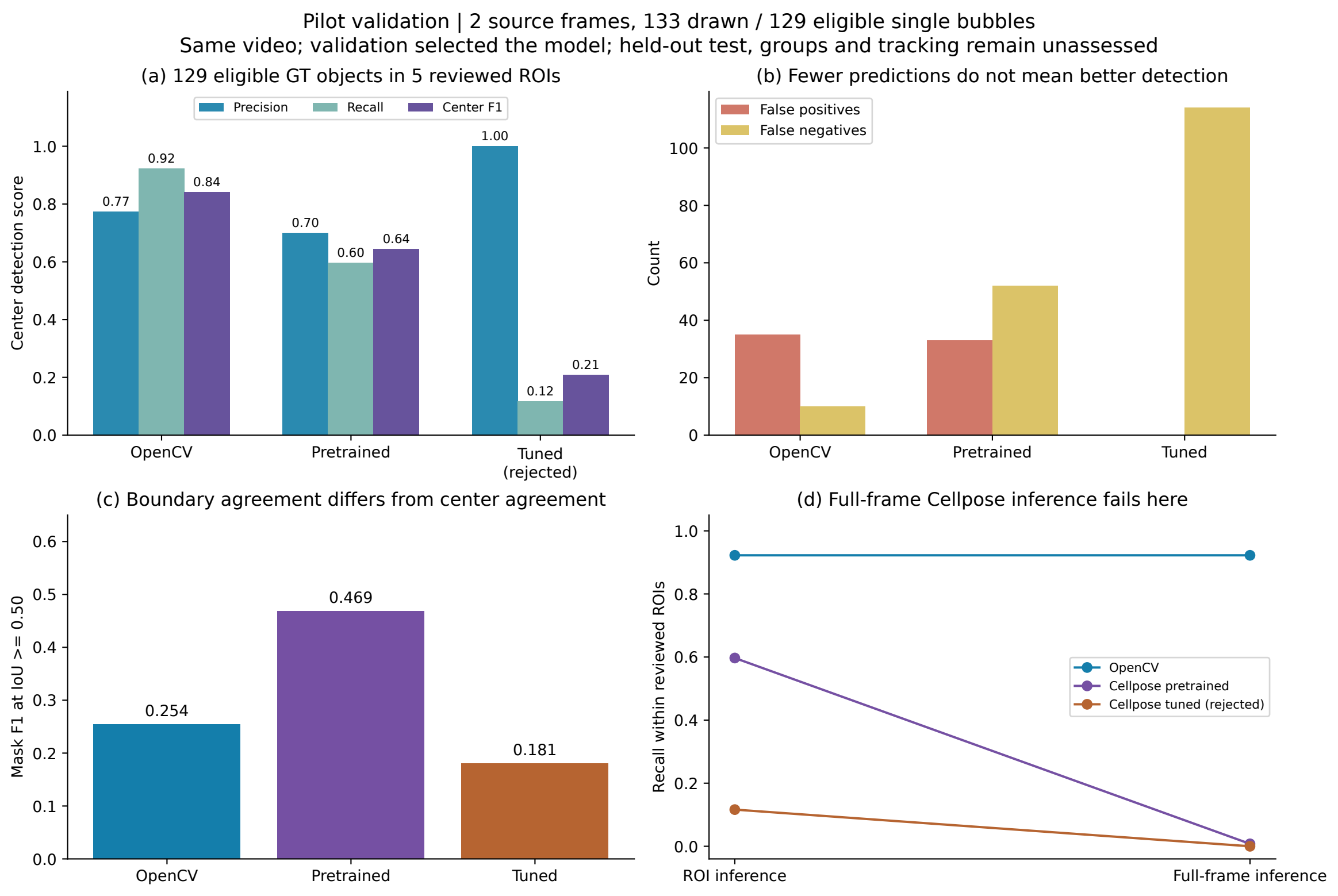
Pilot validation | 2 source frames, 133 drawn / 129 eligible single bubbles
Same video; validation selected the model; held-out test, groups and tracking remain unassessed
(a) 129 eligible GT objects in 5 reviewed ROIs
Precision
Recall
Center F1
Center detection score
0.77
0.92
0.84
0.70
0.60
0.64
1.00
0.12
0.21
OpenCV
Pretrained
Tuned (rejected)
(b) Fewer predictions do not mean better detection
False positives
False negatives
Count
OpenCV
Pretrained
Tuned
(c) Boundary agreement differs from center agreement
Mask F1 at IoU >= 0.50
0.254
0.469
0.181
OpenCV
Pretrained
Tuned
(d) Full-frame Cellpose inference fails here
Recall within reviewed ROIs
OpenCV
Cellpose pretrained
Cellpose tuned (rejected)
ROI inference
Full-frame inference